

Project Design Phase-II

Data Flow Diagram & User Stories

Date	06 July 2026
Team ID	SWTID-2026-5277
Project Name	Food Ordering Behaviour and Consumer Trends
Maximum Marks	4 Marks

Role	Name
Team Leader	Rishika Bhati
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Team Member	Vinita Sharma
Team Member	Chhavi Sharma

Data Flow Diagrams

A Data Flow Diagram (DFD) is a visual representation of how information moves through a system — how data enters, how it is transformed, where it is stored, and how it ultimately reaches its users. For the **Food Ordering Behaviour and Consumer Trends** project, the DFDs below map how raw ordering data (order value, cuisine, meal type, delivery fee, time taken, ratings, mood, hunger level, weather, etc.) flows from customers and delivery partners into a Tableau-driven analytics pipeline, and how the resulting insights are delivered back to the Business Analyst and Operations Manager.

Two levels of DFD are provided:

- **DFD Level 0 (Context Diagram)** — shows the system as a single process and its interactions with all external entities.
- **DFD Level 1** — decomposes the system into its major sub-processes (order capture, preprocessing, analytics, dashboarding, feedback, and delivery-performance monitoring) and the data stores between them.

Scenario Context

Scenario 1 – Rahul Sharma (Business Analyst): Analyses order trends, cuisine preferences and city-wise demand using Tableau dashboards; identifies high-performing segments and tracks

how demand evolves over time.

Scenario 2 – Priya Mehta (Operations Manager): Focuses on delivery performance and operational efficiency by analysing delivery-time distribution, order volumes, and peak-hour workload across cities to optimise resource allocation.

Scenario 3 – Neha Reddy (Customer): Values quick delivery, reasonable pricing and cuisine variety; orders during peak meal times and is more likely to reorder when expectations for speed and service are met.

Figure 1: DFD Level 0 (Context Diagram) — Food Ordering Behaviour & Consumer Trends

DFD Level 0 (Context Diagram) — Food Ordering Behaviour & Consumer Trends

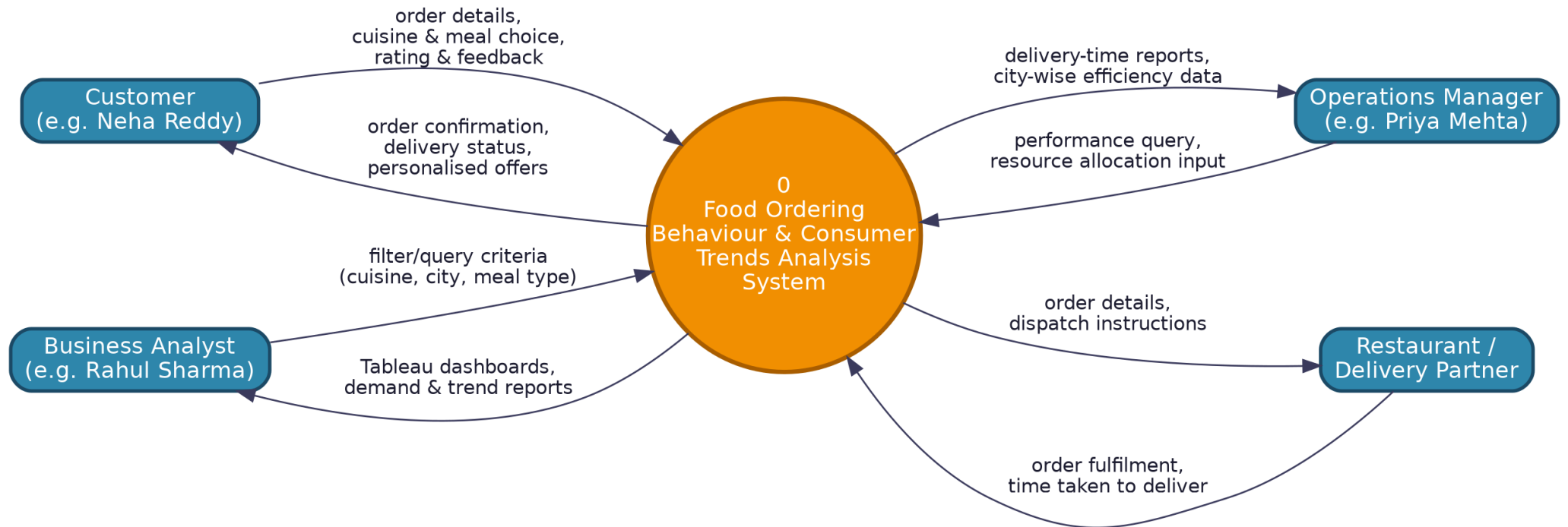
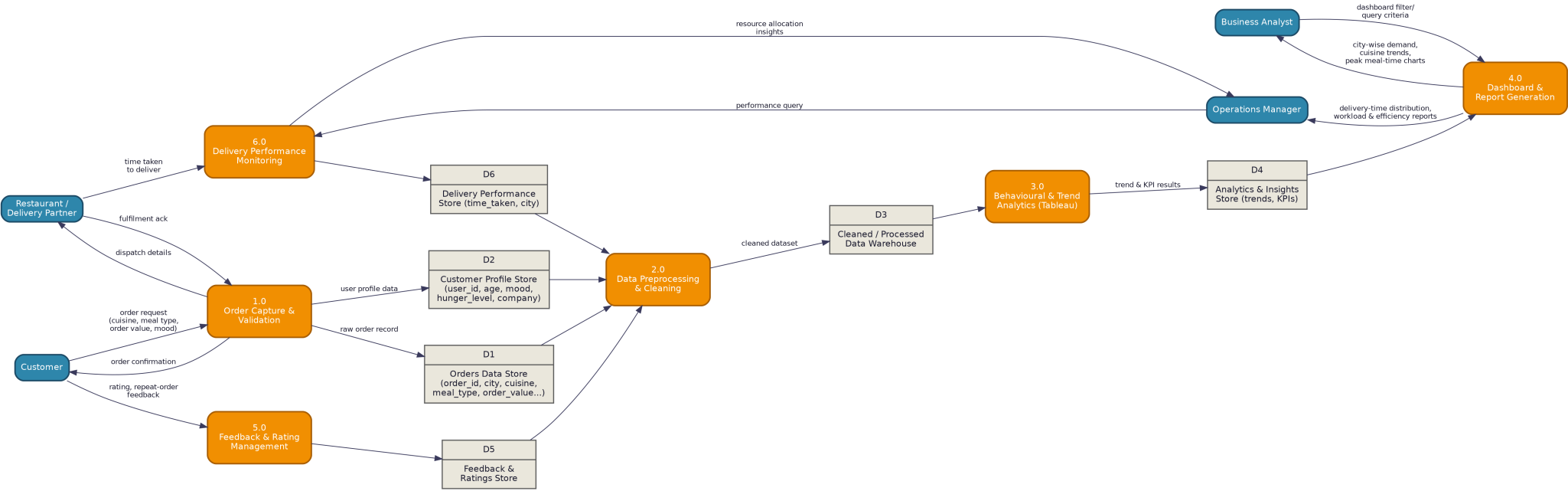


Figure 2: DFD Level 1 — Food Ordering Behaviour & Consumer Trends Analysis System

DFD Level 1 — Food Ordering Behaviour & Consumer Trends Analysis System



User Stories — Food Ordering Behaviour and Consumer Trends

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Customer (Mobile / Web user) [Neha Reddy]	Order Placement	USN-1	As a customer, I can browse and select cuisine (e.g. Chinese, Biryani, South Indian) and meal type (breakfast/lunch/dinner/snacks) so that I can order food matching my mood and hunger level.	Cuisine and meal-type filters are visible and selectable before checkout.	High	Sprint-1
Customer (Mobile / Web user) [Neha Reddy]	Order Placement	USN-2	As a customer, I can view the order value and delivery fee, and apply a discount if available, before confirming my order.	Price breakdown (order value, delivery fee, discount) is displayed pre-checkout.	High	Sprint-1
Customer (Mobile / Web user) [Neha Reddy]	Order Tracking	USN-3	As a customer, I can see the estimated and actual time taken for my order to be delivered so that I know when to expect it.	A live delivery timer/status is shown from order confirmation to drop-off.	Medium	Sprint-1
Customer (Mobile / Web user) [Neha Reddy]	Feedback & Rating	USN-4	As a customer, I can rate my order (1–5) after delivery so that I can share my experience with the platform.	A rating prompt appears once order status changes to delivered; rating is saved.	High	Sprint-2
Customer (Mobile / Web user) [Neha Reddy]	Feedback & Rating	USN-5	As a customer, I can indicate whether I would reorder so that the platform can capture repeat-order behaviour.	"Is repeat order" flag is captured and stored against the order record.	Medium	Sprint-2
Business Analyst [Rahul Sharma]	Order & Demand Trend Analysis	USN-6	As a business analyst, I can view a Tableau dashboard of order volume by city so that I can identify top-performing markets.	Dashboard displays order counts/values grouped by city, refreshable on demand.	High	Sprint-1
Business Analyst [Rahul Sharma]	Order & Demand Trend Analysis	USN-7	As a business analyst, I can filter orders by cuisine and meal type so that I can uncover high-demand segments and customer preferences.	Cuisine and meal-type filters update all dashboard visuals in real time.	High	Sprint-1
Business Analyst [Rahul Sharma]	Spending Pattern Analysis	USN-8	As a business analyst, I can compare order value trends across weekday vs. weekend so that I can identify shifts in customer spending.	Dashboard shows a weekday/weekend comparison view for order value and volume.	Medium	Sprint-2
Business Analyst [Rahul Sharma]	Demand Trend Tracking	USN-9	As a business analyst, I can track how order demand changes across different times of day (morning/afternoon/evening/night) so that I understand evolving customer behaviour.	Time-of-day breakdown chart is available and updates with new data.	Medium	Sprint-2
Operations Manager [Priya Mehta]	Delivery Performance Monitoring	USN-10	As an operations manager, I can view delivery-time distribution by city so that I can identify delays and inefficiencies.	A distribution chart of time_taken_to_order/deliver is available, filterable by city.	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Operations Manager [Priya Mehta]	Resource Allocation	USN-11	As an operations manager, I can monitor peak ordering hours and workload distribution so that I can optimise resource allocation and staffing.	Heat-map/bar chart of order volume by hour and day-type (weekday/weekend) is shown.	High	Sprint-1
Operations Manager [Priya Mehta]	Delivery Performance Monitoring	USN-12	As an operations manager, I can view the impact of rainy weather on delivery time so that I can plan for weather-related delays in advance.	Dashboard segments delivery time by rainy_weather = Yes/No.	Medium	Sprint-2
Administrator (Data/Platform team)	Data Management	USN-13	As an administrator, I can upload and validate the raw food-ordering dataset so that missing or inconsistent values are cleaned before analysis.	Uploaded file is validated; missing/invalid records are flagged before entering the analytics store.	High	Sprint-1
Administrator (Data/Platform team)	Access Control	USN-14	As an administrator, I can manage dashboard access permissions for analysts and managers so that data is only viewed by authorised team members.	Role-based access (Analyst, Ops Manager) restricts dashboard editing vs. viewing rights.	Low	Sprint-2